# Recommender Systems
# lecture 7: bandits for recommender systems

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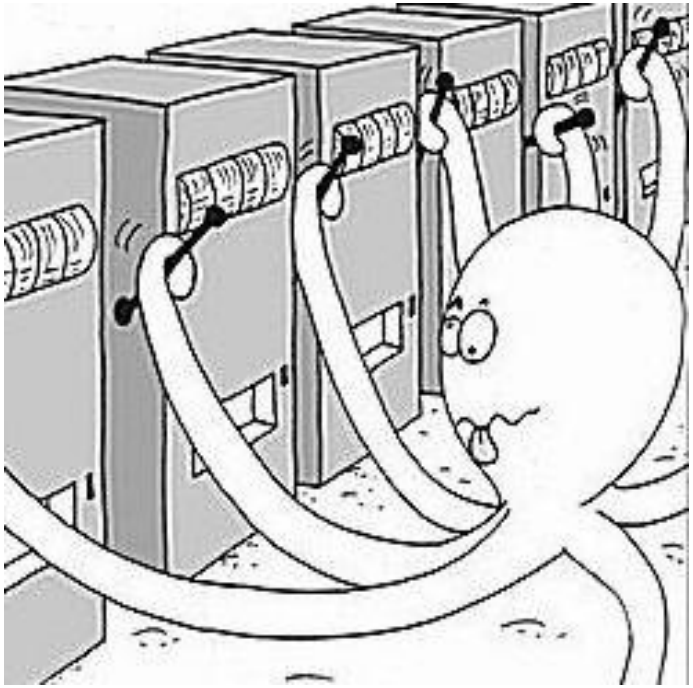
# Multi-armed bandits



**Research question:** how should I allocate my research time amongst my favorite open problems so as to maximize the value of my completed research?

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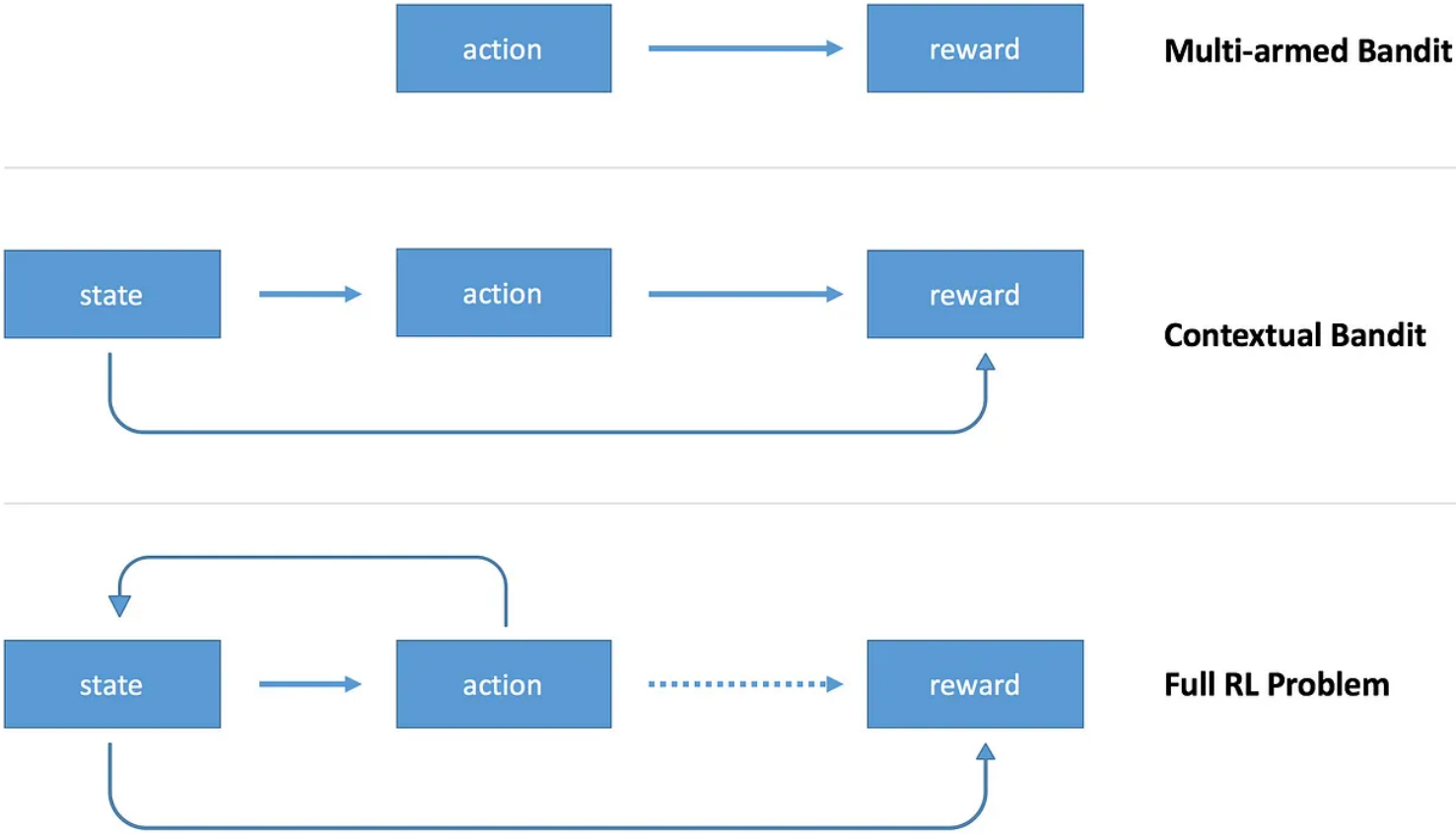
# Bandits overview

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# Classical bandit game (stohastic bandits), Robbins (1952)

**Parameters available to the player:** the number of rounds $n$ and the number of arms $K$.
**Parameters unknown to the player:** the reward distributions $\nu_1, \ldots, \nu_K$ of the arms (with respective means $\mu_1, \ldots, \mu_K$).

For each round $t = 1, 2, \ldots, n$:

1. The player chooses an arm $a_t \in \{1, \ldots, K\}$.
2. The environment draws the reward $r_t$ from $\nu_{a_t}$ (and independently from the past given $a_t$ ).

**Goal:** Maximize (in expectation) the cumulative rewards. Equivalently we want to minimize the cumulative regret:

$$R_n = n\mu^* - \mathbb{E} \sum_{t=1}^{n} r_t,$$

where $\mu^* = \max_{i=1,\ldots,K} \mu_i$

# Applications in recommender systems

## Objective

1. regret minimization
2. BAI: identify the most popular items with fewest possible samples

## Arm

item (e.g. ad, news)

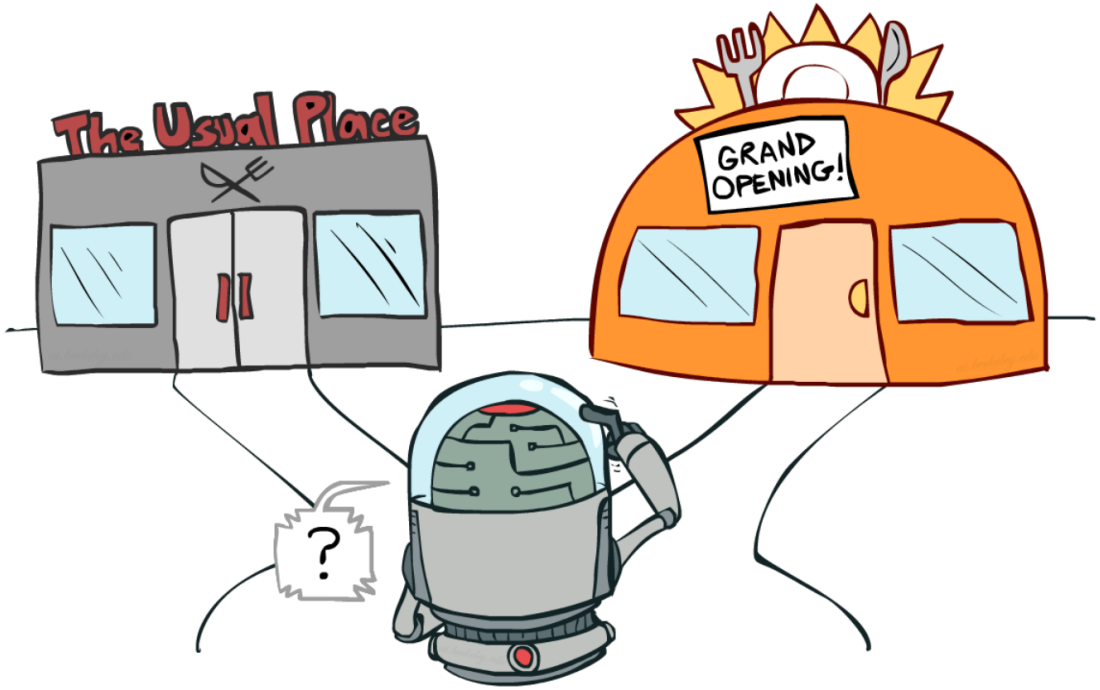
## Reward

click, purchase

## Examples

- Discover new user interests (exploration-exploitation tradeoff)
- reduce model uncertainty in regions of sparse user interaction/feedback
- Select image for film banner

# Exploration vs Exploitation

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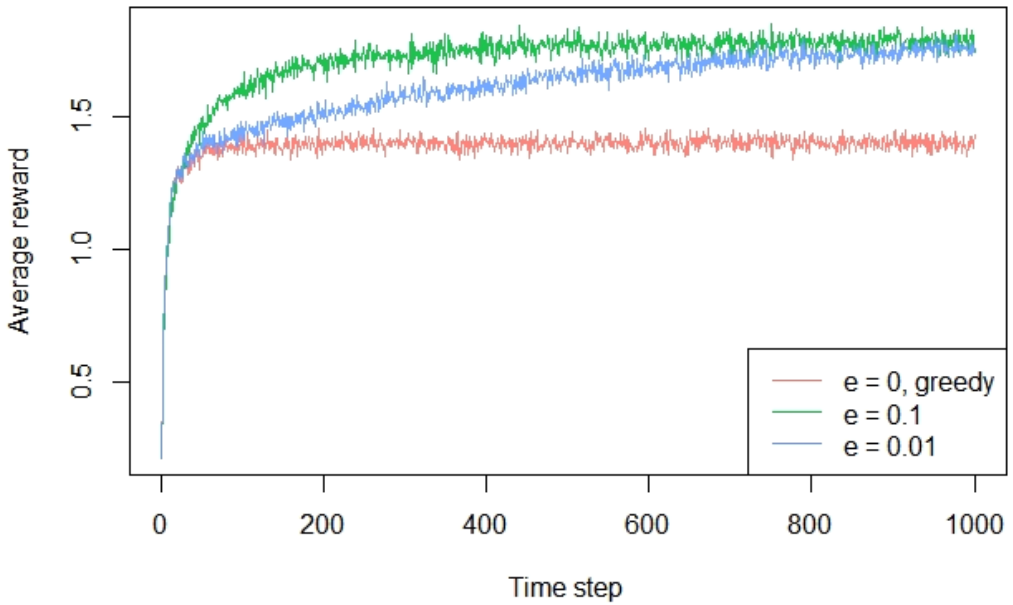
# $\varepsilon$-greedy

$$Q_t(a) = \frac{1}{N_t(a)} \sum_{\tau=1}^{t} r_\tau \mathbb{1}\left[a_\tau = a\right]$$

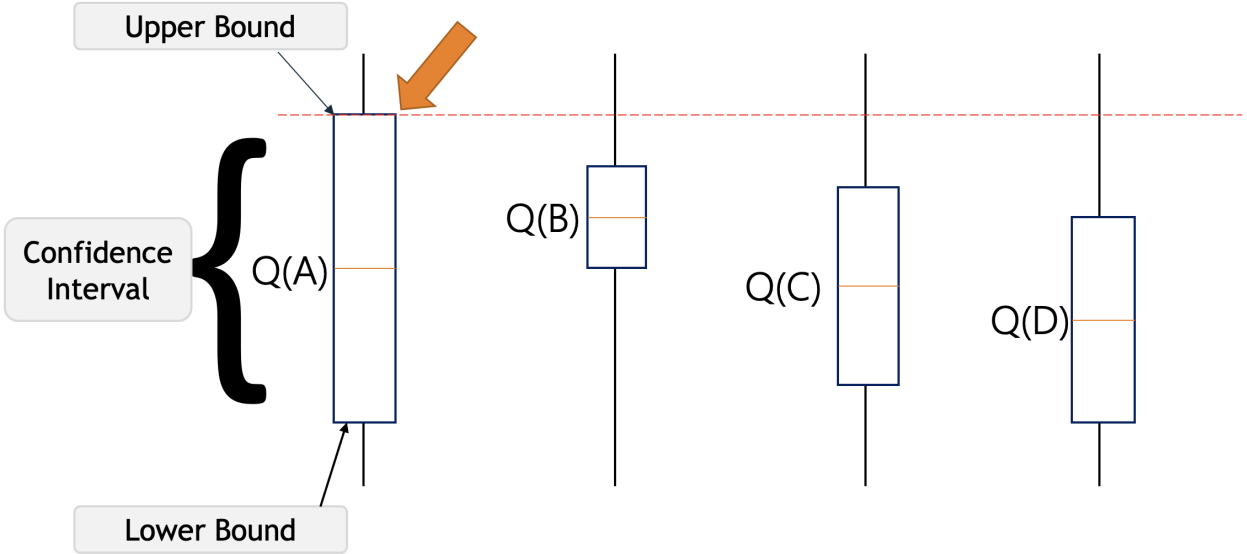
$$a_t = \begin{cases} \arg\max\limits_{a \in \mathcal{A}} Q_t(a), \text{with probability } 1 - \varepsilon \\ \text{random, with probability } \varepsilon \end{cases}$$




Example from Sutton book (source)

# Upper Confidence Bound (UCB)

$$a_t = \arg\max_{a \in \mathcal{A}} Q_t(a) + U_t(a)$$

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# Estimating confidence bounds

## Hoeffding's Inequality

Let $X_1, \ldots, X_t$ be i.i.d. (independent and identically distributed) random variables and they are all bounded by the interval $[0, 1]$. The sample mean is $\bar{X}_t = \frac{1}{t}\sum_{\tau=1}^{t} X_\tau$. Then for $\mathrm{u} > 0$, we have:

$$\mathbb{P}\left[\mathbb{E}[X] > \bar{X}_t + u\right] \leq e^{-2tu^2}$$

$$\mathbb{P}\left[Q(a) > \hat{Q}_t(a) + U_t(a)\right] \leq e^{-2tU_t(a)^2}$$

$$e^{-2tU_t(a)^2} = p \Rightarrow U_t(a) = \sqrt{\frac{-\log p}{2N_t(a)}} \quad (p = t^{-4} \text{ called } \mathrm{UCB}_1)$$

## General UCB formula

$$a_t = \arg\max_{a \in \mathcal{A}} Q(a) + \alpha\sqrt{\frac{\log t}{N_t(a)}}$$

# Thompson Sampling

Set of past observations $D = (a_i, r_i)_{i=1}^N$ modeled with $P(r|a, \theta)$.
Given $p(\theta)$, the posterior distribution is given by the Bayes rule:

$$P(\theta \mid D) \propto \prod P\left(r_i \mid a_i, \theta\right) P(\theta)$$

**Algorithm 2** Thompson sampling for the Bernoulli bandit

**Require:** $\alpha, \beta$ prior parameters of a Beta distribution
  $S_i = 0, F_i = 0, \ \forall i$. {Success and failure counters}
  **for** $t = 1, \ldots, T$ **do**
    **for** $i = 1, \ldots, K$ **do**
      Draw $\theta_i$ according to Beta$(S_i + \alpha, F_i + \beta)$.
    **end for**
    Draw arm $\hat{\imath} = \arg\max_i \theta_i$ and observe reward $r$
    **if** $r = 1$ **then**
      $S_{\hat{\imath}} = S_{\hat{\imath}} + 1$
    **else**
      $F_{\hat{\imath}} = F_{\hat{\imath}} + 1$
    **end if**
  **end for**

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# LinUCB (contextual bandits)

Assumption: reward is linear over state (context)

$$\mathbf{E}\left[r_{t,a} \mid \mathbf{x}_{t,a}\right] = \mathbf{x}_{t,a}^{\top}\boldsymbol{\theta}_a^*$$

**Algorithm 1** LinUCB with disjoint linear models.

Inputs: $\alpha \in \mathbb{R}_+$
**for** $t = 1, 2, 3, \ldots, T$ **do**
  Observe features of all arms $a \in \mathcal{A}_t$: $\mathbf{x}_{t,a} \in \mathbb{R}^d$
  **for all** $a \in \mathcal{A}_t$ **do**
    **if** $a$ is new **then**
      $\mathbf{A}_a \leftarrow \mathbf{I}_d$ ($d$-dimensional identity matrix)
      $\mathbf{b}_a \leftarrow \mathbf{0}_{d\times 1}$ ($d$-dimensional zero vector)
    **end if**
    $\hat{\boldsymbol{\theta}}_a \leftarrow \mathbf{A}_a^{-1}\mathbf{b}_a$
    $p_{t,a} \leftarrow \hat{\boldsymbol{\theta}}_a^{\top}\mathbf{x}_{t,a} + \alpha\sqrt{\mathbf{x}_{t,a}^{\top}\mathbf{A}_a^{-1}\mathbf{x}_{t,a}}$
  **end for**
  Choose arm $a_t = \arg\max_{a\in\mathcal{A}_t} p_{t,a}$ with ties broken arbitrarily, and observe a real-valued payoff $r_t$
  $\mathbf{A}_{a_t} \leftarrow \mathbf{A}_{a_t} + \mathbf{x}_{t,a_t}\mathbf{x}_{t,a_t}^{\top}$
  $\mathbf{b}_{a_t} \leftarrow \mathbf{b}_{a_t} + r_t\mathbf{x}_{t,a_t}$
**end for**

mean (to exploit)

Variance (to explore)

UCB style

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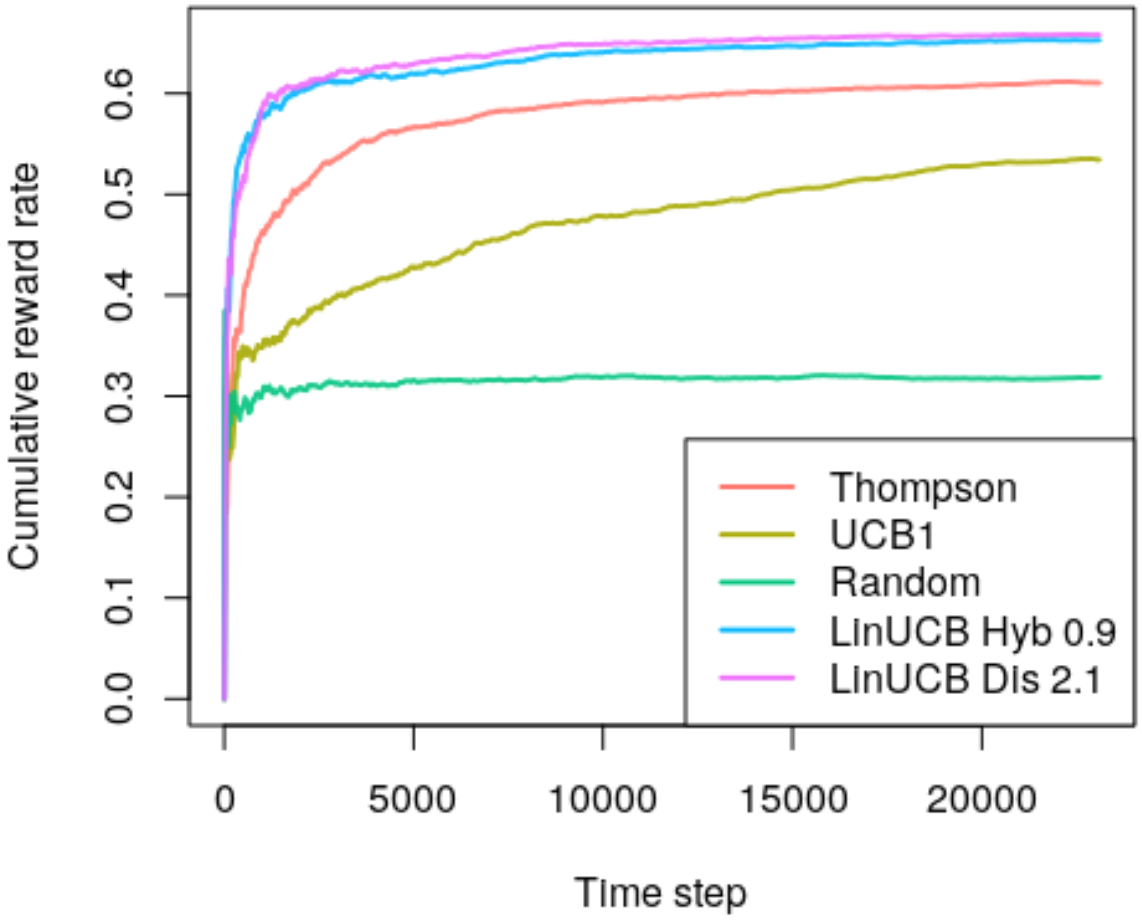
# Algorithms comparison (Movielens-10M)

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# Can you beat the bandit?

1. https://iosband.github.io/2015/07/28/Beat-the-bandit.html
2. http://apbarraza.com/bandits_activity

# Literature


1. *Richard S. Sutton, Andrew G. Barto* (2018). Reinforcement Learning: An Introduction
2. *Sebastien Bubeck and Nicolo' Cesa-Bianchi* (2012). Regret Analysis of Stochastic and Nonstochastic Multi-armed Bandit Problems
3. *O. Chapalle et al.* (2012). An Empirical Evaluation of Thompson Sampling
4. *D. Russo et al.* (2017). A Tutorial on Thompson Sampling
5. *L. Li et al.* (2010). A Contextual-Bandit Approach to Personalized News Article Recommendation